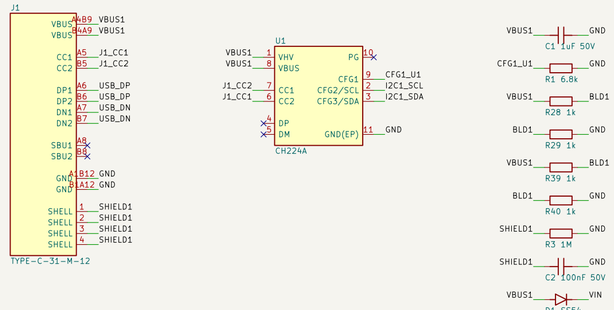
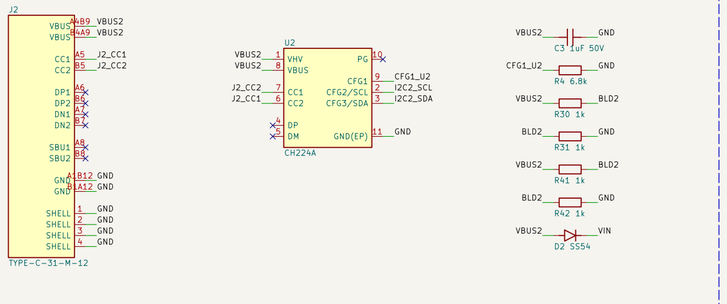
(1) USB-C J1: data + power / U1: PD sink, requests 9V

CH224A: Rset 6.8k on CFG1 = 9V (datasheet table 5-1). VHV and VBUS pins tied to the connector VBUS, 1uF on VHV.
CC1/CC2 go straight to U1 (Rd is inside U1). The connector's CC1 goes to pin 6 and CC2 to pin 7 (crossed) so that the
two tracks do not have to cross on the board; the chip treats both pins alike (plug orientation only).
DP/DM of U1 are not used (data goes to J4 only).
CFG2/CFG3 = I2C (0x22/0x23, address not selectable) -> own bus per chip. PG is not used (state is read over I2C).
D1 blocks back-feed from VIN to the PC. Shield: 1M || 100nF to GND.
R28+R29 || R39+R40 = 1k bleeder (4 x 1k 0805, 110mW each at 21V): pulls VBUS1 below 0.8V against the reverse
leakage of D1 when only J2 is powered.



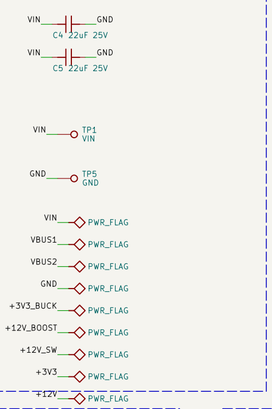
(2) USB-C J2: power only (charger) / U2: PD sink, requests 9V

Same circuit as (1). D+/D- of J2 are not connected. Shield of J2 goes straight to GND.
D2 blocks back-feed from VIN to the charger port: D1/D2 together pick the higher input.



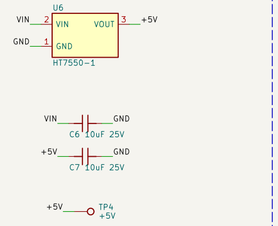
VIN bus (4.2 .. 9V)

No TVS: every part on VIN and on VBUS1/2 stands 21V
(U3 30V, U4 24V, U6 30V, capacitors 25V, CH224A 30V.
bleeders 110mW of 125mW), so a 20V charger fault does
not break anything. Bulk capacitors sit behind D1/D2.



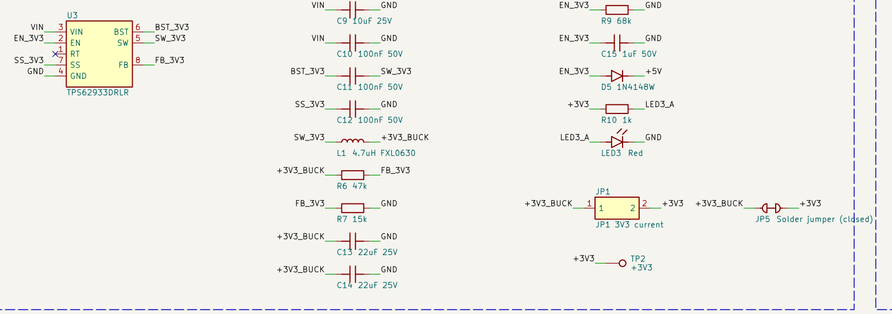
U6: 5V LDO (MCU + J4 VBUS)

HT7550-1: 1=GND 2=VIN(tab) 3=VOUT. 10uF in / 10uF out
(datasheet application circuit).
At VIN=5V the output follows VIN minus dropout
(about 4.2 .. 4.9V).



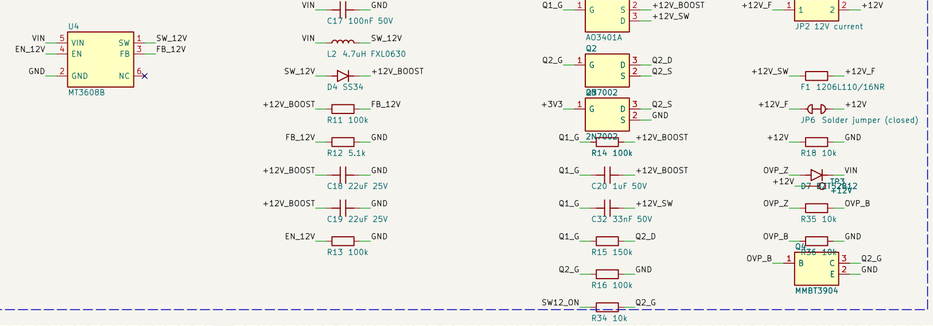
U3: 3.3V / 3A buck (TPS62933) + EN from J1 VBUS + JP1 (current measurement)

Vout = 0.8V x (1 + 47k/15k) = 3.307V. RT open = 500kHz. L = 4.7uH (datasheet table 10-2). SS 100nF -> tSS = 100n x 0.8 / 5.5u = 14.5ms.
EN = VBUS1 x 68k/(100k+68k) = 0.405 x VBUS1: on at VBUS1 > 2.92V typ (3.09V max), off below 2.68V typ (2.51V min). EN = 2.0V @5V, 3.6V @9V.
D5 clamps EN to +5V + VF (< 6V abs max) if VBUS1 ever goes higher. MCU (PC3, low or Hi-Z only) pulls EN low through R38 1k. C15 1uF: EN rises in about 37ms.
Charger on J2 alone -> VBUS1 = 0 -> EN low -> 3.3V stays off.



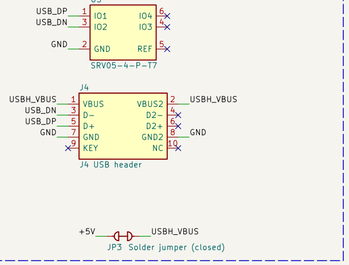
U4: 12V / 0.5A boost (MT3608B) + Q1/Q2 output switch + JP2 (current measurement)

Vout = 0.6V x (1 + 100k/5.1k) = 12.36V. L = 4.7uH. D4 = SS34 (datasheet application). EN pulled down 100k -> off with a blank MCU.
A boost passes VIN to its output when stopped, so Q1 (AO3401A, Vgs max +-12V) cuts the output. Q2 and Q3 in series pull the gate down
through R15 150k only when SW12_ON is high AND +3.3V is present: Vgs = -12.36 x 100k/250k = -4.9V (-5.5V at 13.8V).
C20 1uF gate-source holds the gate when the charger is plugged in (C32 would otherwise pull it on); C32 33nF drain-gate
sets the edge to about 1.8 V/ms (inrush 0.2 .. 0.4A into 100 .. 220uF). Q2 gate pulled down 100k -> off with a blank MCU.
F1 (PTC 1.1A hold, 0.66A at 70C) protects against a short on the card. R18 bleeds the +12V line to 0V when off.
Over-voltage lock: when VIN > about 13V (12.2 .. 13.6V), D7 + Q4 pull the gate of Q2 low, so Q1 cannot turn on (9V PD: always on, 15V: always off).



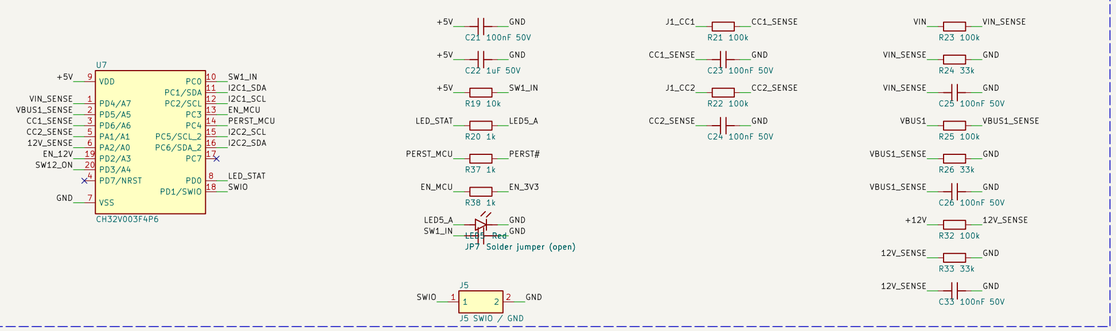
USB 2.0 data: J1 -> U5 (ESD) -> J4

SRV05-4: 1=IO1 2=GND 3=IO2 4=IO3 5=REF 6=IO4.
D+ on IO1, D- on IO2 (one diode pair per line).
IO3, IO4 not used.
REF is left open: its TVS breaks down at 6V.
VBUS1 can be 9V.
J4 = motherboard USB 2.0 header pinout:
1 VBUS, 3 D-, 5 D+, 7 GND, 9 key (pull the pin).
JP3 (solder jumper, closed) feeds +5V to J4 pin 1.



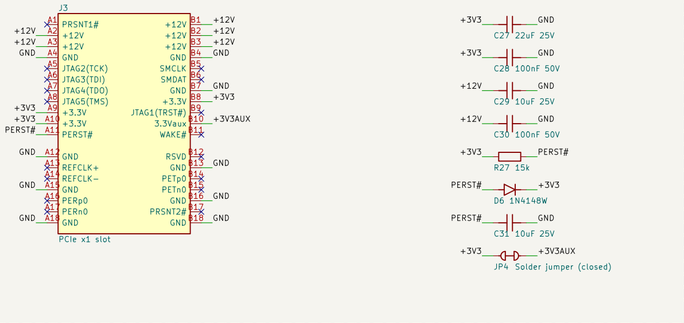
U7: CH32V003F4P6 — reads CC / VIN / VBUS1, switches 3.3V and 12V

VDD = +5V (4.2 .. 5V). ADC reference = VDD, so firmware corrects with the internal Vref channel.
CC1/CC2: 100k series + 100nF (the 100k keeps the load off the PD line: fault current into the pin < 0.2mA).
VIN / VBUS1 sense = x 33k/(100k+33k) = 0.248: 5V -> 1.24V, 9V -> 2.23V, 20V (TVS clamp) -> 4.96V (below VDD = 5V whenever VIN > 5.5V).
PC3 = EN_3V3 and PC4 = PERST# through 1k (R38, R37): drive low or leave Hi-Z, never push-pull high. PD2 = EN_12V, PD3 = SW12_ON (100k pull-downs).
PA2/A0 = +12V sense (x 0.248: 12.36V -> 3.07V).
I2C1 (PC1/PC2) -> U1. Remap I2C1 to PC5/PC6 (AFIO I2C1_RM = 10b) -> U2. No external pull-ups: CH224A has them inside.
J5: 1 = SWIO, 2 = GND (WCH-LinkE). JP7: solder closed = manual 12V override (replaces the slide switch).



J3: PCI Express x1 slot — power pins only

Pin names follow the PCIe CEM pinout (side B = component side of the card, side A = solder side).
+12V: A2 A3 B1 B2 B3. +3.3V: A9 A10 B8. 3.3Vaux: B10 via JP4 (solder jumper, closed).
GND: A4 A12 A15 A18 B4 B7 B13 B16 B18. PERST# (A11): 15k to +3.3V, 10uF to GND -> 2.0V about 140ms after 3.3V. D6 empties C31 when 3.3V drops.
PRSNT#, WAKE#, JTAG, SMBus, REFCLK and the lane pins are left open.



Mechanical

M3 mounting holes (not plated,
no net).



Generated by scripts/gen_schematic.py — edit the script, not this file
USB-C x2 -> 3.3V / 12V for USB-data PCIe tuner cards (PX-MLT5PE etc.)

danything

Sheet: /
File: pcie-tuner-board.kicad_sch

Title: PCIe tuner external power board

Size: A2	Date: 2026-11-12	**Rev: C**
KiCad E.D.A. 10.0.8		Id: 1/1